Assignment 2

Student Name: Kanishk Bhandari | UID: 23BCS11100
Branch: BE CSE | Section/Group: 23BCS_KRG-2_B
Semester: 6th | Date of Submission: 02/03/2026
Subject Name: System Design | Subject Code: 23CSH-314

1- **Aim** - To design a scalable and highly available YouTube-like distributed video streaming system for large-scale users.

2- **Requirements**: Functional & Non-Functional

A- Functional Requirement

- User Registration & Login
- Video Upload by Creator
- Search Videos
- Watch Videos in multiple resolutions (360p, 720p, 1080p, 4K)
- Like a Video
- Comment on a Video
- View Like count & Comments

B- Non-Functional Requirement

- Scalability: 500M+ Users
- High Availability >>> Consistency
- Low Latency (<100ms interactions)
- Zero / Negligible Buffering
- High write throughput for Likes & Comments

3- **Core-entities of System**

- Users
- User Metadata
- Videos
- Video Metadata
- Manifest File
- Video Segments
- Likes
- Comments

4- **API endpoint creation**

1. **Video Upload**
   1. POST Call: /video/upload
   2. POST Call: /video/{video_id}/metadata

2. **Streaming**

   1. GET Call: /video/{video_id}
   2. GET Call: /video/{video_id}/manifest.mpd
   3. GET Call: /video/{video_id}/manifest.m3u8
   4. GET Call: /video/{video_id}/segment/{segment_id}

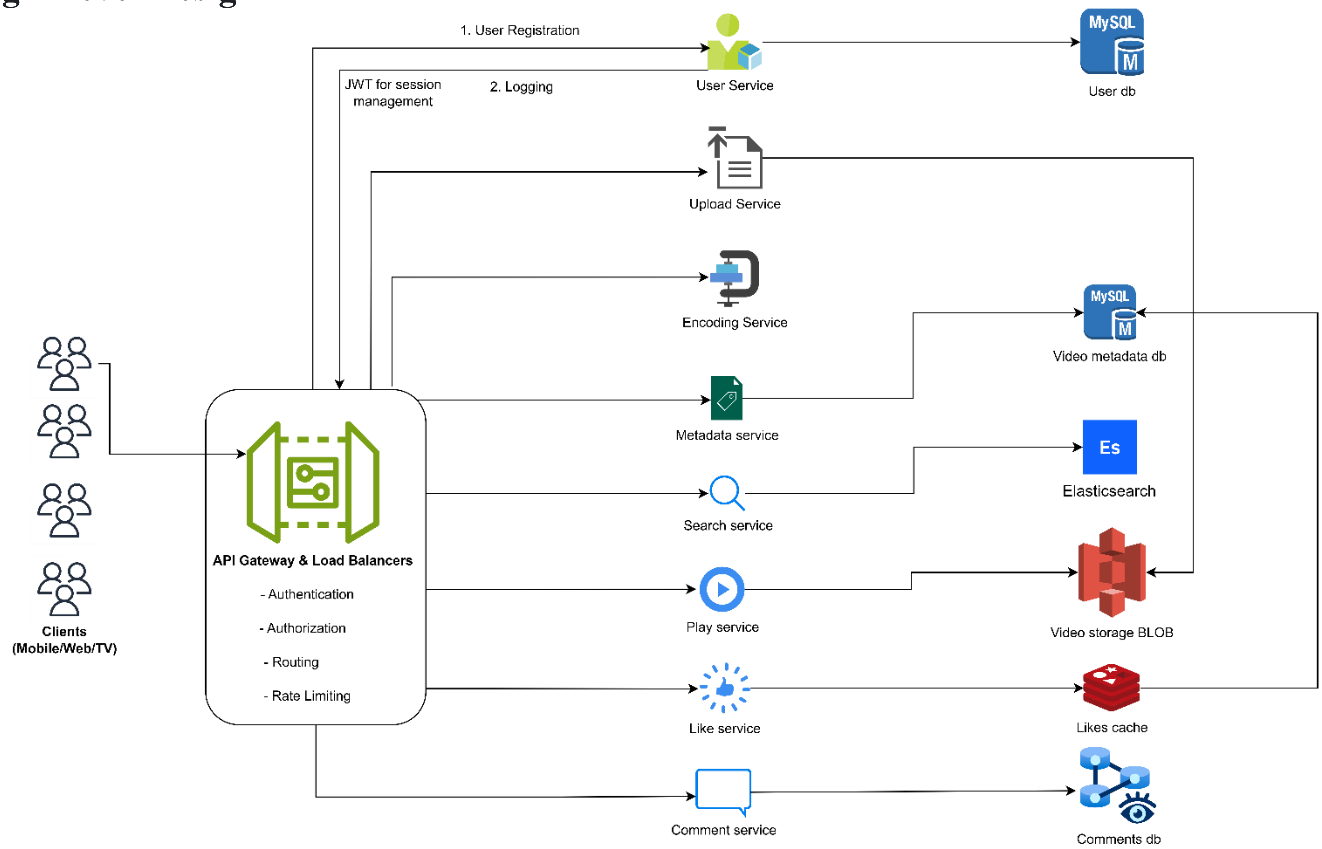
3. **Likes**

   1. POST Call: /video/{video_id}/like
   2. DELETE Call: /video/{video_id}/like
   3. GET Call: /video/{video_id}/likes/count

4. **Comments**

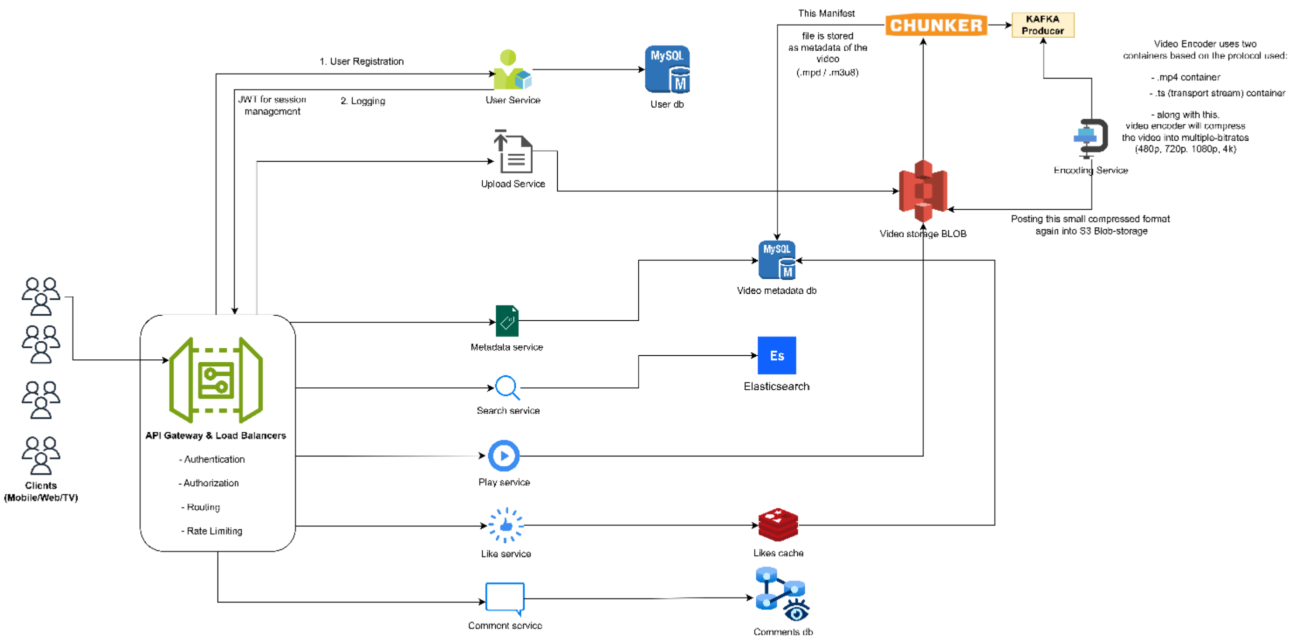
   1. POST Call: /video/{video_id}/comment
   2. GET Call: /video/{video_id}/comments?cursor=xyz

5- High-Level Design

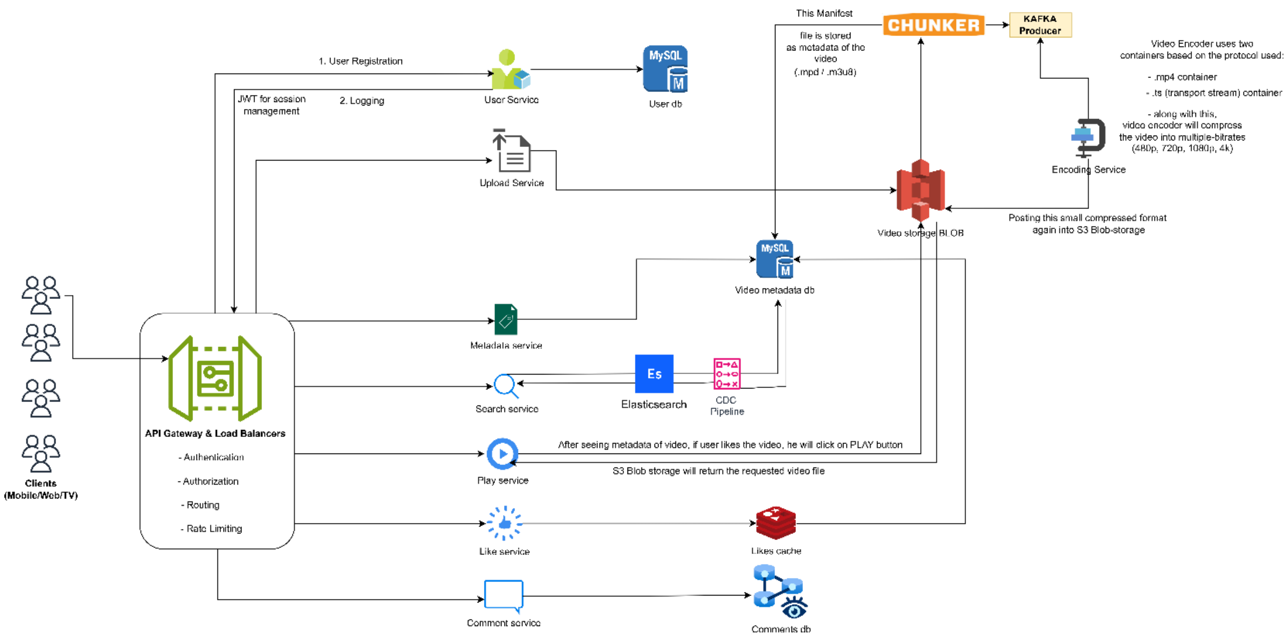


6- Low-Level Design

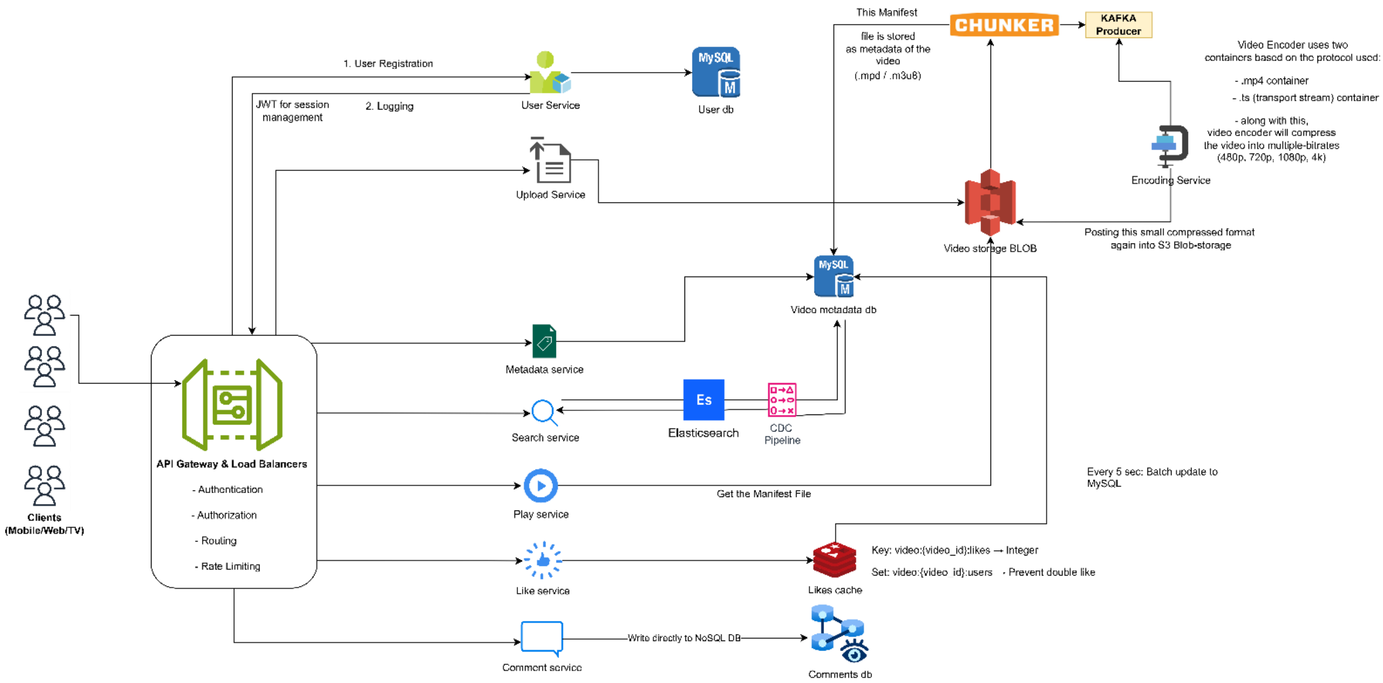
Upload Service



Search Service



Like and Comment Service



Problem: Heavy Video Traffic
Solution: CDN, Edge caching, Horizontal scaling